The Action of the Alternating Group A_5 on the Leech Lattice via the Conway Group Co_0

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Abstract

The Leech lattice Λ_{24} is the unique even unimodular lattice in $\mathbb{R}^{24}$ with no vectors of squared norm 2. Its automorphism group is the Conway group Co_0, a finite group of order 8 315 553 613 086 720 000. In this paper we study the natural orthogonal action of the alternating group A_5 (of order 60) as a subgroup of Co_0. We review the background on the Leech lattice and Co_0, exhibit an embedding of A_5 into Co_0, describe the restriction of the 24-dimensional representation, and examine the fixed-point sublattices stabilized by A_5. Particular emphasis is placed on the classification of the 290 Co_0-orbits of fixed-point sublattices due to Höhn and Mason, in which several orbits have pointwise stabilizer isomorphic to A_5 or its double cover $2.A_5$. We discuss the resulting coinvariant lattices and the geometric interaction between the group action and the lattice structure.

1 Introduction

The Leech lattice Λ_{24} occupies a central place in mathematics, appearing in sphere packing, coding theory, finite simple groups, and moonshine phenomena. Discovered by Leech in 1965, it is the unique even unimodular lattice in Euclidean 24-space with no roots (vectors of squared norm 2). Its full group of isometries fixing the origin is the Conway group

$$\mathrm{Co}_0 = \mathrm{Aut}(\Lambda_{24}),$$

a sporadic group containing the three Conway simple groups Co_1, Co_2, and Co_3 as quotients or subgroups.

The alternating group $A_5 \cong \mathrm{PSL}(2,5)$ is the smallest non-abelian simple group. While the root lattice A_5 cannot embed isometrically into the rootless Leech lattice (as it contains norm-2 vectors), the *group* A_5 does embed faithfully into Co_0. This embedding realizes a faithful orthogonal action of A_5 on Λ_{24} by lattice-preserving isometries.

In this expository paper we describe this action, its construction, and its interaction with the lattice geometry. We rely on the classification of fixed-point sublattices by Höhn and Mason (2015), which shows that A_5 arises as a pointwise stabilizer for several Co_0-orbits of sublattices of Λ_{24}. These fixed-point sublattices provide a concrete geometric picture of the representation theory of A_5 inside Co_0.

2 Background on the Leech Lattice and Conway Group

2.1 The Leech Lattice

The Leech lattice Λ_{24} may be constructed in several equivalent ways. One standard construction uses the binary Golay code $\mathcal{C}_{24}$ of length 24:

$$\Lambda_{24} = \left\{ (x_1, \ldots, x_{24}) \in \mathbb{R}^{24} \;\middle|\; 2x_i \in \mathbb{Z},\ \sum x_i \equiv 0 \pmod 2,\ \text{support of } (x_i \bmod 2) \in \mathcal{C}_{24} \right\},$$

equipped with the Euclidean inner product scaled so that the minimal nonzero norm is 4. It is even ($\langle v, v\rangle \in 2\mathbb{Z}$ for all $v \in \Lambda_{24}$) and unimodular ($\Lambda_{24}^* = \Lambda_{24}$).

Key properties:

- No vectors of norm 2.
- 196 560 vectors of norm 4 (the *minimal vectors*).
- The automorphism group Co_0 acts transitively on these minimal vectors.

2.2 The Conway Group Co_0

The group Co_0 is the full orthogonal group of Λ_{24}:

$$\mathrm{Co}_0 = \{g \in O(24, \mathbb{R}) \mid g(\Lambda_{24}) = \Lambda_{24}\}.$$

It has a center $\{\pm I\}$ of order 2, and the quotient $\mathrm{Co}_1 = \mathrm{Co}_0/\{\pm I\}$ is simple. The order is

$$|\mathrm{Co}_0| = 2^{22} \cdot 3^9 \cdot 5^4 \cdot 7^2 \cdot 11 \cdot 13 \cdot 23.$$

Co_0 contains many interesting subgroups, including extensions involving alternating groups $2.A_n$.

3 The Alternating Group A_5

The alternating group A_5 is the group of even permutations on 5 letters. It is simple of order $60 = 2^2 \cdot 3 \cdot 5$ and isomorphic to the icosahedral rotation group. Its irreducible representations over $\mathbb{C}$ have dimensions $1, 3, 3, 4, 5$.

In the context of Co_0, we consider the restriction of the natural 24-dimensional real orthogonal representation of Co_0 to a copy of A_5. Because A_5 is simple and non-abelian, any homomorphism $A_5 \to \mathrm{Co}_0$ is injective (the kernel is normal and proper, hence trivial).

4 Embedding A_5 into Co_0

Several constructions realize $A_5 < \mathrm{Co}_0$:

1. **Monomial and octonionic constructions.** Wilson (using octonions) exhibits generators of a copy of $2.A_5$ inside Co_0 via signed-permutation matrices and monomial transformations preserving the Leech lattice. Extending by an outer automorphism yields A_5 itself.

2. **Subgroup chains.** The ATLAS of Finite Groups and the maximal-subgroup tables of Co_1 show that A_5 appears inside larger subgroups such as $(2.A_5 \circ 2.\mathrm{HJ}) : 2$, where HJ is the Hall–Janko sporadic group. Quotienting by the center produces an embedding of A_5 into $\mathrm{Co}_1 \hookrightarrow \mathrm{Co}_0$.

3. **Fixed-point stabilizers.** As classified by Höhn and Mason, several Co_0-orbits of fixed-point sublattices have pointwise stabilizer exactly A_5 (or $2.A_5$).

The action is by orthogonal transformations: each $g \in A_5 \subset \mathrm{Co}_0$ satisfies $g^T g = I$ and $g(\Lambda_{24}) = \Lambda_{24}$.

5 The 24-Dimensional Representation Restricted to A_5

The natural module $\mathbb{R}^{24}$ decomposes under A_5 into irreducible representations. Since the full representation of Co_0 is irreducible over $\mathbb{R}$, the restriction to A_5 is a faithful orthogonal representation of dimension 24. Typical decompositions (computed via character tables) involve multiples of the 4- and 5-dimensional irreps of A_5, consistent with the known character table.

On the lattice level, the action preserves the integral structure. For a generator $\sigma \in A_5$ of order 5, the fixed-point sublattice Λ^σ has rank equal to the multiplicity of the trivial representation in the restriction.

6 Fixed-Point Sublattices and the Höhn–Mason Classification

A fundamental geometric object associated with the action is the *fixed-point sublattice* of a subgroup $G \leq \mathrm{Co}_0$:

$$\Lambda^G = \{v \in \Lambda_{24} \mid g \cdot v = v \text{ for all } g \in G\}.$$

The orthogonal complement (coinvariant lattice) is

$$\Lambda_G = (\Lambda^G)^\perp \cap \Lambda_{24}.$$

Höhn and Mason (arXiv:1505.06420) classified all 290 Co_0-orbits of such fixed-point sublattices, tabulating for each orbit:

- Rank of Λ^G,
- Discriminant form and genus,
- Stabilizing subgroup G,
- Root system (if any) of Λ^G and Λ_G.

Several orbits have pointwise stabilizer isomorphic to A_5 (order 60) or its double cover $2.A_5$. For example:

- Orbits with $\mathrm{rk}(\Lambda^G) = 5$ or 4, where the coinvariant lattice has rank 19 or 20.
- In these cases the induced action of A_5 on Λ_G is faithful and often irreducible or decomposes into known A_5-modules.

These fixed sublattices are self-normalizing in Co_0 and provide explicit models of the representation theory inside the lattice.

The interaction is mutual: the lattice constrains the possible group actions (no norm-2 vectors can be fixed in certain ways), while the group symmetries illuminate the deep hole structure and deep-hole diagrams of Λ_{24}.

7 Interactions via Group Action

The action of A_5 on Λ_{24} preserves:

- The quadratic form $\langle \cdot, \cdot \rangle$,
- The set of minimal vectors (though not necessarily transitively),
- The Golay-code structure in coordinate descriptions.

Conversely, the lattice geometry forces the representation to be orthogonal and integral. Fixed-point sublattices give rise to interesting K3 surfaces and modular forms via the associated vertex operator algebras (in the context of moonshine).

No non-trivial element of A_5 can fix a norm-4 vector (by simplicity and the transitivity properties of Co_0), which is consistent with the classification.

8 Conclusion

The embedding of A_5 into Co_0 exemplifies the rich interplay between finite simple groups and exceptional lattices. While the root lattice A_5 cannot embed into Λ_{24}, the group A_5 acts faithfully and provides a beautiful example of how sporadic symmetry groups manifest geometrically through fixed-point sublattices. The Höhn–Mason classification offers a comprehensive catalog for further study of these interactions, with potential applications to vertex operator algebras, K3 surfaces, and modular moonshine.

Future work could include explicit matrix generators for an A_5-copy inside Co_0 or computations of the associated modular forms attached to the coinvariant lattices.

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